

Problem

What is the phase sequence of a balanced three-phase circuit for which

$$\mathbf{V}_{an} = 120\angle 30^\circ \text{ V} \text{ and } \mathbf{V}_{cn} = 120\angle -90^\circ \text{ V} \text{ ? Find } \mathbf{V}_{bn}.$$

Step-by-step solution

Step 1 of 1

Consider the following data:

$$\mathbf{V}_{an} = 120\angle 30^\circ \text{ V}$$

$$\mathbf{V}_{cn} = 120\angle -90^\circ$$

From this

$$\mathbf{V}_{cn} = \mathbf{V}_{an}\angle -120^\circ \text{ V}$$

'C' phase lags 'A' phase by 120° . So this is an acb phase sequence.

For acb phase sequence, calculate the voltage $\mathbf{V}_{bn}$.

$$\begin{aligned}\mathbf{V}_{bn} &= \mathbf{V}_{an}\angle 120^\circ \text{ V} \\ &= 120\angle (30^\circ + 120^\circ) \\ &= 120\angle 150^\circ \text{ V}\end{aligned}$$

Therefore, the voltage $\mathbf{V}_{bn}$ is $\boxed{120\angle 150^\circ \text{ V}}$.